Hear thee hear thee. Below is a list of Edicts for teams to follow in addition to the DTEG Revision 4 from the 2025-2026 competition cycle and whichever is the most current revision to the R&R. The DTEG is long and needs an overhaul in terms of language. The expected timeline for DTEG revisions pushes past the entry window into late October if not mid November. Rather than have teams start a project and then have to comply with a new DTEG after they start the design, we have created a list of edicts for teams to follow to correct issues we have noticed in lacking elements of the DTEG and elements that teams clearly are not reading in the current DTEG that we need to strengthen the language on. Most of these are already in the DTEG and need to be emphasized here. Teams for Launch Canada 2027 will need to follow the DTEG and this list of edicts that will make it into R5 of the LC DTEG to be released in a couple months. The final wording from these edicts may change by release of the new DTEG but the rules herein and intent will not. Follow the letter and spirit of these edicts and the R4 DTEG second and you should be compliant with the R5 DTEG. You still have to be compliant with the R&R revision that is currently active at any time. For clarification, the progress report system is changing its name to the range safety report.

https://docs.google.com/document/d/19R6k41PHZqsZfoFB0OJ5mRpOpO8Djx6CSxiqebDQJwE/edit?usp=sharing

All Rockets

Stability

- Length to Diameter Ratio
    - The L:D Ratio shall NOT exceed 45 and it should NOT exceed 25.
    - Range Safety highly recommends no rocket above a LD Ratio of 25 if the team expects a stable flight.
- Damping ratio
    - The Damping Ratio of any rocket must be between 0.5 and 0.03 it should be between 0.3 to 0.05.
    - Range safety highly recommends no rocket should have a damping ratio above 0.3 or below 0.05 to prevent extreme weather cocking and failure to recover from oscillations.
- Static Margin
    - Minimal Flight Stability of 10% of Body Length throughout the flight from rail exit to twice the burn duration of a stage into flight
        - 10%BL = 1 caliber / LD ratio of 10 (sound familiar? 1 caliber stability rule? Yeah thats what its based on)
        - 1.5 Caliber / LD 25 => 6% %BL
        - 1.5 Caliber / LD 50 => 3% %BL

        - 2.5 Caliber / LD 25 => 10% %BL
        - Using % of Body Length is a more appropriate unit for long rockets like the ones that come to competition.

Electronics

- Altimeter bays for any stage traveling over 0.7M must be concentric and have concentric air sampling holes. Any crescent av bays must be in rf transparent material
- No more than one altimeter per circuit
- No custom circuitry or software used for arming of a COTS altimeter
- SRAD PCBs used with commercial altimeters can only be used for simplifying the wiring harnessing of a single COTS board.
- Hardware shunts for are required for sustainer ignition circuits
- Physical switch required for all electronics. Any wireless switch is required to be in series with a physical switch to be used to keep avionics disarmed while raising and lowering the tower if the switch is not accessible by an outstretched arm of your second tallest team member attending the competition. The team is responsible for securing access to a ladder tall enough to properly reach any switch where they do not climb high enough that their knees exceed the top rung of the ladder. If a LC ladder is not available that is tall enough, then your team has to secure a ladder elsewhere or build a shorter rocket.
    - Three points of contact must be maintained on a ladder at all times with the climbers' shoes secured to their feet. Untied shoes or flip flops are not allowed on the ladder.
- For All Lithium Chemistry Batteries - Vinyl or shrink wrap for fire and puncture protection
    - Do not ziptie a lipo pouch to a plank of wood or plastic
    - Cells need padding
- Individual power supplies are required for each individual deployment altimeter and each individual GPS or RF beacon.
    - The Launch window is roughly 8 hrs. Typically you are not armed for that long. You might be armed for at most 2 hours at a time. External charging allows you to top up and reduces on-vehicle battery mass and therefore lithium mass and battery size. You are launching into a forest. You need to limit the amount of lithium on board your aircraft. You should seek on-rail charging of each individual power supply. When using minimal GSE or LC launch control for a solid rocket motor where you don’t build your own GSE you should build in a connection for charging individual batteries plugged into an portable external power supply that can be removed before flight.

Structures

- Radax joint must be mechanically pinned and chemically bonded not either
- Recessed nozzles must account for an expansion fan of at least 45 degrees without conflicting with the airframe
- Building Materials - 3d printed stuff
    - Pre sliced (in the cad) holes for tapping or bolting into. Do not drill into the infill or insert heat set inserts into the infill.
    - There will be more specific rules for acceptable filaments and slicer settings for flight hardware. In the meantime have fun prototyping. Need multiple walls and quasi isotropic infill like gyroid or cubic probably want between 5-25% after that you don't get much more strength. Increase the perimeter or wall count to strengthen the part.
- No metal rail buttons. Machine them out of Delrin (Polyoxymethelene, POM Acetal)

Operations

Teams need to prioritize ease of integration on launch day and transfer between setup area and pads. Practice assembly and disassembly and design for its ease.

Assembly Dress Rehearsal - Strongly Suggested for Single Stage Solids Teams Required for all Advanced Rockets

- Assembly of the rocket from the state it will be packed in
- Integration of recovery system
- Fire drogue ejection charge - pop test
- Show the drogue comes out
- For single bay dual deploy perform a yank test to show the main parachute doesn't come out
- Fire main ejection charge or release (and yank the main in the case of a single bay deployment) to show it does come out when it is supposed to.
- This test must be done with a completed rocket as it is going to be at competition with no change to nose weight or tolerancing of tubes or additional or removal of shear pins
- You need a pop test on all components that are designed to separate in flight. Every main bay and drogue bay and separation system is to be tested even if they are identical from another stage or as a duplicate of the previous release (ie drogue and main)

Rules and Requirements Timeline and Misc.

- Your Final Design Report shall be on a finished rocket or test stand.

- Your Final Range Safety Report is on a finished rocket / test stand / project. It shall include documentation and images of the final rocket / test stand next to the design cad, sims etc.
- Range Safety Report (RSR) and Range Safety Deadlines - Exact dates will be finalized with 2027 R&R likely mid October.
    - 1st RSR With IDP November
    - 2nd RSR End of Feb
        - Site and Launch Control Layout Due - for ADV and Static
    - 3rd week of June, Engine tests deadline - you need a week to write the range safety report with this data.
    - 4th week of june, Final RSR End of June
        - The rocket is done. With the exception of paint it is done. Nothing is left to drill or assemble.
        - Wet Dress Rehearsal including the abort test is done
        - Flight Dynamics are done with real masses and thrust curves
        - Vehicle Integration Rehearsal is done
        - Final roster of how many people, which people, which subteam leads, tshirt sizes, are done
        - Engine tests are done
        - Avionics, electronics, controls, etc have all wiring properly run
        - Recovery System is fully assembled and fit tested
        - You should be able to take the rocket in this state to competition and fly it
        - After this date your team is to run drills to practice assembly and operation
    - (1st - 2nd week of July) reviewers are reviewing (teams need to sleep and rehearse integration of their test vehicle alongside prepping for exams)
    - (3rd - 4th weeks of July) are Virtual Inspections. (teams need to sleep and rehearse integration of their test vehicle alongside prepping for exams)
    - First week of August is RS prep for departure
    - Second week August is site setup
    - Third week August is competition
- Mentors need to be hands off. Teams are the ones fabricating, testing, and integrating the rocket. A mentor is someone that has experience in what the team is trying to build that the team can confide in and feels comfortable failing in front of. Their job is to advise and refer to other techniques and processes that might not have been noticed by the team. This person must be nurturing but pushes them to do better. This is the person they turn to for counsel and not judgment. LC will provide plenty of judgment along side design reviews that really should be analyzed by people outside of the team's history where their experience and opinions are not entrenched in the way it has always been done there.

Frequency Allocations

So LC is getting rather large which is a good thing. We are however limited to the maximum number of frequencies active at the same time. The following may seem draconian but it is the only way to make the math work. Final site development at Advanced will be 20 Paddocks. As of the 2025 competition entry I knew of just over 20 hybrid and Liquid Projects in Canada. Many of the advanced projects skip the LC tech dev phase so there are likely more that will pop up for 2027. Some were still in the static fire phase but matriculation to competition in the advanced launch category has increased from 50% to 66% for hybrids and liquids flights. Advanced as a whole has gone up to 77%. In the next two years depending on trends in matriculation and growth in the category I expect to fill all 20 paddocks, which is why I have been so focused on site clearing. When prototyping a band plan I have to make the assumption we are getting to that full plateau. In prototyping the band planning using what I put together for tracking after virtual inspection, this is what is clearly available: 440 MHz band is so empty that with 125kHz channel widths, we can accommodate each adv Hybrid, Liquid, and Tribrid team having a 440 MHz transceiver like the Ham shield or an alternate for data and command transmission on top of enough tracking for every last team to have assigned tracking frequencies through the competition. For the purposes of longevity these frequencies will still be assigned when teams move to the pad after inspection.
Teams can run cables for GSE but that is a nightmare and routing lengths at Advanced need to be established to avoid the roads. It would likely max out at 3500-4000 feet. 1500 ft would be the minimum for only one team and it is hard to know which that will be until after all Virtual Inspections. I've done that over better terrain and it's awful but doable. Light based communications or wireless laser communications would not be constricted by the number of teams.

Whether the restrictions will have to be in place or not will only be determinable after virtual Inspection. We will have some idea where the problems will lie after the first range safety report, but we won't have any idea on matriculation until virtual inspection. Given the expected numbers I have to make the assumption that you might all choose the same band and remain as bandwidth intensive as y'all already have been. Wide band transmissions cannot be allowed at Advanced as there is no way to get that math to work. 20 MHz band widths make 2.4 and 5.8 so crowded that reservation of nearly the entire band for Drone and livestream ops seems like a better use of the bandwidth. 10 MHz band width in 2.4 and 5.8 are to be expected but still likely having restrictions. Teams need to reduce the amount of data they need to transmit over these frequencies. Teams need to transmit state detection signals (Pressure, Mass, Temperature, Valve position detection, etc) as numerical data and not as a video of a gauge or a valve. The teams that are not able to restrict usage of the 2.4 and 5.8 to narrow bandwidths will face reduced salvo opportunities.

Avenues being explored are similar to the voting methods and rail sharing regulations already in use for which teams would be allowed to operate in a salvo that had more demand than supply of channels. That decision hasn't been made yet and won't be by DTEG release. So it will be written as the whim of the Advanced Launch Manager. It will likely be one of the proceedings as it is less intensive to manage. It is possible that by some horrible perfect storm of low matriculation of dissimilar systems that everyone gets their own channel and there is no problem but soon we cannot rely on teams not making it to competition and the band won't be crowded. There is also an equally likely scenario where only teams that chose one band made it and it is crowded but the others are vacant. The fact of the matter is that it is currently crowded and even if there was perfect coordination we are reaching the point that we would max out the sum total of frequencies in 2.4 and 5.8. I would recommend that if you have a simple system you need to minimize the data transmission that your setup requires and stick to 440 MHz. 915 MHz has an issue with GPS demand overlap. That will only be fixed by widescale adoption of the Featherweight swift, BRB900L, and BRB433L trackers. This will be expensive along with purchase of 6 GHz links and in the future there is the potential for a new 7 GHz band.

- Frequencies for GSE will be assigned after virtual inspection but before arrival to competition
- Frequencies for tracking will be assigned by LC when a team passes Final Range Safety **AND** is sent to a pad or is added to a salvo at Advanced.
    - Teams not in a salvo or passing range safety will be required to be set to a designated Setup Area Frequency as to not conflict with any system being used by a rocket preparing to fly
- No more than x2 frequency allocations per stage on board a rocket between the 220 MHz, 440 MHz and 915 MHz bands.
    - This is for all purposes and is not limited to the mandatory 1 frequency per stage for GPS tracking.
    - This includes any SRAD data transmission and the mandatory GPS system
    - Exception in the 915 MHz band if the only 915 MHz transmitters are the Featherweight Swift GPS or BRB900L , 2 are allowed per stage but no other 915 Transmitter is allowed in GSE, SRAD, GPS or otherwise when a 915 tracker is being used.

220 MHz

The 220 MHz band is perfect for RDF beacons as an alternative to GPS. There is enough bandwidth that we will only limit one 220 MHz beacon per stage. Systems include but are not limited to ham shields, BRB beacons, ElvinC / rocket beacons.

- Only one 220 MHz beacon is allowed per stage

- Bandwidth is restricted to 25 kHz

440 MHz

The 440 MHz Band is nearly empty and there are products that allow data packet transmission in this amateur band with the easiest license to obtain. BRB433L is highly recommended to prevent future conflicts as with time slots and selected channels it creates an option for 960 channels.

- No more than one 440 MHz transmission per stage
    - An exception is the BRB433 L with no more than two 440 MHz per stage. Cannot be paired with the broadband 440 transmitters.
- One 440 MHz channel with 300kHz width will be made available per static fire, hybrid launch, liquid launch, or tribrid launch team for GSE.
- Teams will need to be able to transmit within 125KHz (250 kHz range) around a center, provided by LC.

915 MHz

Due to the prolific nature of the featherweight GPS and a few other 915 trackers, restrictions on the 915 MHz band are necessary. There are 28 Channels (with 2 time moderated subchannels) for the classic featherweight GPS. Some of this spectrum has to be dedicated to GSE, SRAD Avionics, and Fluctus systems. Fluctus uses twice as much bandwidth per module. At LC 2026 advanced teams used 3.25 channels per stage in 440 MHz and 915MHz. 915MHz was favored 2:1 over 440 MHz. If this continued by LC 27 there likely would be no available spectrum in 915 MHz for Restricted, Junior, Cert or Sport flights. We have to curtail this usage in two ways: 1) Cap the number of transmitters per project. 2) Incentivize the selection of narrow bandwidth GPS systems. 3) Only assign frequencies when the project is about to fly and shut it down as soon as it lands. Featherweight, a brand that many of you rely on, has a new tracker that costs as much as the old one but has much more narrow bandwidth and has auto frequency hopping. Big Red Bee has a BRB900L that has a similar configurable lora transmitter that can reduce bandwidth. We need teams to start utilizing the Swift and the BRB900L tracker this year. If each rocket stage of every last team that applied last year went to competition as a flight project (so what we could reasonably expect next year) used a swift tracker or BRB900L we would use 120 of 240 channels in 915 MHz as opposed to 120 of the 56 available. The math doesn't work for 915 MHz without it and / or a significant shift to 440 MHz and 220 MHz beacons.

- one 915MHz transmission per project including GPS and GSE (Two stages only get one)
    - Exception for 915 MHz: teams are allowed to use up to 2 featherweight swift or BRB900L trackers per stage in lieu of any other 915 transmitter on the rocket or ground. Original Featherweight trackers cannot be used in conjunction with Swift

Trackers. Ex) a two stage team can have 4 swift trackers or 4 BRB900Ls or 1 Original Featherweight GPS or Fluctus or SRAD or 915 GSE. A single stage team can have one Original Featherweight or Fluctus Tracker or SRADs or 2 Swift Trackers or 2 BRB900Ls. Teams can even have a mix of Swifts and BRB900Ls

- In the crowded 915 band teams operating a SRAD tracking or GSE control system will have to broadcast in 915 with 200kHz (400KHz range) around a center freq provided by LC.
- 915 MHz allocations for GSE are subject to denial from a salvo based off of available channels at the time of the salvo and the selection of which teams are approved for operating in those bands is to be managed by the Advanced Launch Manager. Allocation method will likely be first come first serve but is subject to change by the ALM. No more than (x10 tentatively) 400kHz width Channels will be made available per salvo for GSE.

Drone, GSE, and Video Streaming Frequencies

5.8 GHz and 2.4 GHz telemetry are subject to denial from salvo based off of available band at the time of the salvo and the selection of which teams are approved for operating in those bands is to be managed by the Advanced Launch Manager. Allocation method will likely be first come first serve but is subject to change by the ALM. All 2.4 GHz and 5.8 GHz communication must be done with tight beam antennas and narrow spectrums. Priority for the 2.4 GHz and 5.8GHz Spectrum is as Follows:

Range Clearing Drone
LC Livestream Video Connection
LC Livestream Drones
Teams in a Salvo
Drone Operations of teams in the Salvo
Non Launching Team Testing Operations
Non Launching Teams and Personal Drone Operation

Drone operation will be managed by drone control in the 2.4 GHz and 5.8 GHz spectrums. This will take up a huge amount of the spectrum. Likely at least half if not more. As teams use more drones, drone usage may be restricted or GSE usage of the 2.4 GHz and 5.8 GHz spectrums.

2.4 GHz

No more than 40 MHz or 2.4 GHz will be available for use by GSEs. With 10 MHz width channels that means no more than 4 will be operated simultaneously by teams. Paddocks will be given a channel and a polarization of the directional antenna. If the polarization does not work onsite launch capability cannot be expanded to the potential 8 (or fewer) available channels and orientations. This needs to be tested on site setup day, no testing would be

possible by LC until then. Relying solely on 2.4 GHz is a risk. The orientation of your antenna will not be determined until after Virtual Inspection but before arrival at competition. Mounting hardware should be made for both orientations at each end. Channel selection must be manual and should be confined to 10 MHz bandwidth.

5.8 GHz

No more than 50 MHz of 5.8 GHz possibly only 40 MHz will be available for use by GSE. With 10 MHz channels No more than (5) 10MHz width 5.8 GHz channels will be operated simultaneously by teams. Relying solely on 5.8 GHz is a risk. Paddocks will be given a channel and a polarization of the directional antenna. If the polarization does not work onsite launch capability will not be expanded to the potential 10 (or fewer) available channels and orientations. This needs to be tested on site setup day. The orientation of your antenna will not be determined until after Virtual Inspection but before arrival at competition. Mounting hardware should be made for both orientations at each end. Channel selection must be manual and should be confined to 10 MHz bandwidth.

6 GHz

This is a newly available option. There are channels that are bonded for higher bandwidth. There are the following options with mixed bandwidths. Each would be controlled based on demand.
(x29) 40 MHz channels (3, 11, 19, 27, 35, 43, 51, 59, 67, 75, 83, 91, 99, 107, 115, 123, 131, 139, 147, 155, 163, 171, 179, 187, 195, 203, 211, 219, 227)
(x14) 80MHz channels (7, 23, 39, 55, 71, 87, 103, 119, 135, 151, 167, 183, 199, 215)
(x7) 160MHz channels (15, 47, 79, 111, 143, 175, 207)
(x3) 320MHz channels (31, 95, 159)

60 GHz

No more than (5) 60 GHz 2.16 GHz width channels will be operated simultaneously by teams. Paddocks will be given a channel and a polarization of the directional antenna. If the polarization does not work onsite launch capability will not be expanded to the 10 available channels and orientations. This needs to be tested on site setup day. The orientation of your antenna will not be determined until after Virtual Inspection but before arrival at competition. Mounting hardware should be made for both orientations at each end. Channel selection will be manual and confined to 2.16 GHz bandwidth.

Alternatives to Crowded Frequencies

Teams should consider purchase or development of Wireless Laser Communications, LiFi, or Directable Visible Light Communication. Another option is a server connected to starlink or alternative satellite internet.

Frequency Management

Frequency management is being expanded from Trackers to trackers and GSE to deal with the significant increase in interference in Drone and GSE control. A Band management system is in development for competition but is not ready yet. It likely will be more outlined in the DTEG R5 release.

Advanced Specific

The Minimum TWR for Hybrid, Liquid, and Tribrid teams increases to 8 for LC 2027. The Minimum TWR for Hybrid, Liquid, and Tribrid teams increases to 10 for LC 2028. The minimum TWR for two stages remains 10.

Pressure Vessels

- **Burst pressures calculated by the material used at the weakest point utilizing the minimal literature value for that temper and alloy**
- MEOP is your pressure relief valve set pressure
- Proof pressure must be **>=**1.5 MEOP
- Burst Pressure must designed to be **>=** 2 MEOP * (appropriate Knock down factor of at least 1.2 for fully penetrated welds only)
    - Weld knockdown factor must be given in the Range Safety Report for the type of weld, material, type of joint, weld quality, and source of the justification for the knockdown factor. No knockdown factor will be allowed lower than 1.2. If the context of the weld requires a knockdown factor greater than 1.2 then you must have a MEOP that is the [burst pressure]/ 2 / [the appropriate knockdown factor]

- COPVs (including combustion chambers) must have a designed burst pressure >= 4*MEOP. COTS tanks don't have to have a proof press if operating below ¼ burst pressure
    - Either batch test showing failure above 4*MEOP or a 1.5 MEOP Proof Press with a calculated burst pressure greater than 4 * MEOP

Testing Requirements

- We require two full duration engine tests for SRAD engines:
    - A low pressure test and a high pressure test to bound the nominal operating range.
        - Teams should also do a test at the desired pressure but bounding it is more important to us.
    - Alternatively two tests near the same pressure and you have to operate at that pressure at competition.
    - Teams must Hydrotest tanks and valves **prior** to cold flows and valve actuation testing respectively. Once those tests are done the abort test must be conducted. Then the cold flow test of the injectors and then hotfire. **Do not hydrotest after the hot fire**. Do not test the valves after the cold flow or abort tests. You need to know the safing systems work and are reliable before you need them.
    - The 20 valve actuations test at MEOP with a simulated fluid or the intended fluid must be done **before** your abort tests, cold flows, and hot fires.
    - The manual abort needs to be tested under pressure (MEOP) but not necessarily liquid load
    - Valve testing needs to consider humidity and cryo conditions. Piloted valves need to be monitored in testing for leakage between the working fluid and actuated flow.
- Clusters need to demonstrate successful simultaneous ignition
    - Simultaneous igniter firing
    - For clustered hybrids/liquids/ tribrids simultaneous successful ignition on at least a partial fill.

Fill to Fire Times

Tanks under 24 L GLPP F2F times must be under 20 Minutes and should be under 10. Abort times must be under 10 minutes and should be under 5

Tanks over 24 L GLPP F2F times must be under 30 Minutes and should be under 15. Abort times must be under 15 minutes and should be under 8

Advanced Probationary Levels

There will be three Probationary Levels (PL-n) within Advanced. Rather than total Impulse, they are governed by the maximum allowable total volume of **Gaseous and Liquid Pressurants and Propellants (GLPP)** to minimize the potential yield of a catastrophic failure. This includes ullage, pressurant, liquid and gaseous oxidizer, and liquid and gaseous fuel. It does not include solid fuels. The volumes are set such that good performance matches the existing framework of impulses. Rather than the team over-building a tank and "whoops" into something larger, if the team extracts more performance from a finite unyielding volumetric limit, then they have done a good job and are allotted that extra impulse, giving them more altitude and velocity. They are also set to push teams to develop higher efficiency projects that have margin in performance so that a lower performance firing at competition can still yield a partially successful flight. Once a team passes a probationary level they are free to operate in the next so long as the team does not fail to successfully launch and successfully recover a rocket at a competition with time constraints and working around other teams in that range of volumes for two years. By the second season without a successful flight and recovery they will be dropped down by at least 1 probationary level. Probation level stands until a rocket is successfully recovered and reaches an altitude over 50% of the target altitude and no more than 25% over the predicted altitude with an **Altitude Adjusted Specific Impulse (AASI)** greater than the level's isp target requirement. Overshooting represents a significant risk to the competition as a whole for our highest altitude flights which your team is likely working towards. All teams are assumed to be in PL 1 until evidence is provided by the team that they meet a higher probationary limit via Competition (time-pressure) flights at this competition or an equivalent one. Teams must supply evidence of current effective probationary level by the first Range Safety Report. Engine test stand data showing total impulse, propellant and pressurant masses for that test, and the target altitude and actual altitude attained will be required. This will be matched with our records and verified with other competitions. In the event that a deviation request for an intermediate scale launch was granted by range safety via a firing during the year yielding a conditional approval the team must have an unarguably successful flight by the final design report otherwise the team is required to obey the original probationary level at competition.

The altitude adjusted isp is calculated as follows

$$Altitude\ Adjusted\ Specific\ Impulse\ (AASI) = min(1, \frac{[actual\ altitude]}{[target\ altitude]}) \times [Static\ Fire\ ISP\ [s]]$$

Probationary Level 1

- **(2 - 8 Liters of GLPP, 15 kNs for Two Stage and Clusters)**
- This is a teams first entry to Complex, Hybrid, Liquid, and Tribrid Rockets in the Advanced Category. For hybrid, liquid, and tribrid rockets we highly recommend a 2 Liter entry to spend limited funds on the fabrication and testing of a GSE system and extensive testing campaigns. This will keep the test cost low giving you more tests per dollar building your team's understanding of your system. Continued use of this engine after your competition year will allow for subscale testing of methods of increasing efficiency without larger more costly tests.

- Hybrid, Liquid, and Tribrid Engines must perform in their static fires at a **minimum ISP of 100s** to come to competition. The AASI must exceed 100s to remove PL 1 probationary limit and build an engine with a tank with greater than 8 Liters of GLPP.
- Mixed Propulsion Type Two stage teams will not be allowed while on PL 1 for either their first two-stage or Hybrid, Liquid, or Tribrid engine.

Probationary Level 2

- **(8 - 20 Liters of GLPP, 40 kNs for Two Stage and Clusters)**
- Hybrid, Liquid, and Tribrid Engines must perform in their static fires at a **min ISP of 100s** to come to competition. The AASI must exceed 145s to remove PL 2 probationary limit and build an engine with tank volume greater than 20 Liters of GLPP.

Probationary Level 3

- **(20 - 32 Liters of GLPP)**
- Hybrid, Liquid, and Tribrid Engines must perform in their static fires at a **min ISP 145s** to come to competition. The AASI must exceed 170s to remove PL 3 probationary limit and build an engine with greater than 32 Liters of GLPP.

Cleared Advanced Teams

Non Probationary Teams coming to competition with a GLPP greater than 32 L must exceed 170s in their static fires to come to competition. Teams with AASI lower than 170s from the flight will be considered an unsuccessful firing and can result in a team being put back on probation if they are unsuccessful in flight or recovery the next year.

Advanced Operations

General Operations

- Teams need to use pendants for paddock specific hazard flagging.
- Teams need to use the Leis.
    - It is how we determine who is who and from which subteam when we have a specific technical question
    - It shows who is allowed at the pad under red flag

    - It highlights where your leads are
- Teams need to understand that yellow flag
    - It starts when the votes are counted 1 hr before scheduled salvo starts. That time was voted on by your Points of Contacts.
    - Yellow flag lasts only 50 minutes including transit time
    - No admittance into pads upon yellow flag without final safety inspection done and approved
    - The rocket needs to be passed inspection prior to yellow flag to be in the salvo
    - The rocket should already be on the rail by the time the yellow flag begins. You shouldn't vote for being in that salvo if you aren't already on the rail.
- Teams need to understand Red Flag
    - Red flag operations only last 10 minutes before the salvo is supposed to start
    - Red flag begins 10 minutes before the salvo time your team voted on
    - At Red flag start (except for the maximum of five arming personnel) the rest of the team is at the launch control point
    - At salvo start your entire team needs to be back at their launch control points.
- You need to plan and rehearse your ops to fall within this timeline
    - Wet dress rehearsal needs to be conducted under same time constraints.
- The above time line has not changed. you need to follow it.
- Delays past scheduled salvo start are indicative of poor operations and should be scored accordingly. Now for the stick.
    - We will not tolerate salvo delay past 25 min from the start of a salvo
    - No warning at 0 over. if it is a matter of a couple minutes it doesn't matter enough to rush you. The first vote deduction (halving of the vote) occurs at 1s past salvo start where the team is still at the Pads.
    - A verbal warning will be issued at 10 min over along with the second halving of the vote value.
    - at 20 minutes over there another verbal warning and another third halving of the vote value.
    - We will not allow a cumulative 50 min delay or any delay longer than 25 min in any given salvo as that is creating an overheating issue for electronics, bottles, and people.
    - Every 10 minutes past salvo start is a vote deduction issued at 1s (V/2), 10 minutes (V/4), 20 minutes(V/8), 25 minutes (V/16), 25 min +1 s (V*0)
- Timeout condition
    - At 25 minutes over a team must stop work and retreat to launch control
    - Teams will lose all remaining vote value if they are 25minutes+ over
    - If a team when told at 25 minutes over that they are done do not immediately stop work and remove themselves from the pad they will not fly for the remainder of the competition
        - Their bottles will be closed, igniters removed, power disconnected by LC staff. Then reported to Judging.

        - Your Operation Score will be Zero
        - Your Flight Score will be Zero
    - Any further attempts to launch in that salvo or for the rest of the competition will disqualify the team from reentry into the competition the next year.
    - The team will be expected to break down their rail after the last salvo of the day

Voting

- Advanced teams still have two infinitely divisible votes.
    - The first vote is with a weight of 1
    - the second vote has a weight of ½
    - The third vote has a weight of ¼
    - The nth vote has a weight of 2^{n-1}
- The Point of contact is a single individual that will be summoned and extracted from their duties on the team often
    - They are the ones that vote
    - They must have a full high level understanding of the entire project
    - The subteam leads need to keep the point of contacts apprised of any developments or delays so they have full situational awareness and can make decisions quickly with the other points of contact without the need for receiving updates prior to a decision
- Votes will be granted to competing teams and non-competiting teams.
    - Teams with more than one project from the same institution will have their votes split between the two projects / teams.
    - Each project requires a different point of contact
- Teams can only vote for or against launch salvos AFTER they've completed final rocket review the day of launch - unless they have sequestered their rocket overnight (in which case the approval from the day before holds).
- Vote value resets to zero once the rocket leaves the rail under thrust. A Reflight attempt would be conducted with no vote value at the convenience of the remaining teams. A team using an LC rail for a reflight attempt is permanently at the back of the line for rail usage if there are non-launched teams that are ready to be railed for a salvo. Any remaining time from the allotted 50 minutes maximum available is not reset to 50 minutes. Any relaunching team will trigger the timeout condition after 10 minutes (normally 25 minutes for a first flight team) delay after salvo start.

Potential New Advanced Area Dependant on Terraforming Operations over the Summer

FAST TO FIRE (FTF) Advanced

- New Area for Advanced likely starting LC 2027
- Must use the ~18' LC 1515 rail without exception all other advanced projects with their own rail or not meeting the F2F requirements for FTF will have to operate off the Advanced Plateau (Current Advanced Area).
- 1 hr ops cycle at this pad so launch at 5 minute to the top of the hour so 9:55, 10:55, 11:55 etc
- 1-3 pads available per hourly salvo
- The team must demonstrate in the dress rehearsal full pads ops under 30 minutes for setup and arming, fill to fire times must be under 5 minutes with a less than 2 minute abort.
- Failure to meet the timing standard in rehearsal will put your team at the Advanced Plateau. Failure to meet the timing standard at competition will result in transfer to the plateau and probation from the FTF area for one year. Teams that have not competed in Advanced before are not eligible for the FTF area.
- Hybrid, Liquid, and Tribrid Teams qualifying for the FTF area will be required to use a 1-2 bottle nitrous or nitrous & pressurant system utilizing a bottle dolley or cart with their entire GSE integrated onto the cart such that the team can load/offload the dolley(s) or cart(s) on/off their truck with only four people in under a minute. The sum total of GSE prep would be to offload the dolley/cart, roll it to the pad hook up to the rocket, plug into an LC Fire Control, and open the bottle(s). The battery should already be hooked up. You will connect your valve control, load cell and pressure transducers to the LC fire control. This is possible to do, it requires planning, testing, and rehearsal. It is as hard as your team makes it. Some designs that have been made will not meet this standard. This standard is necessary to meet a timeline that allows for rapid salvos where voting is not necessary. If you don't meet this standard you will fly from the traditional advanced area.
- Efforts to meet this standard will improve your operational outcomes at the traditional advanced area even if you miss this standard. If you wish not to be inconvenienced by other teams your team must be itself: convenient. This is the standard to aim for. If your team and other teams attempt to meet this standard, then conditions will improve even for those that don't meet the standard.
- Example Timeline
    - 10:15 Yellow Flag setup begins
    - 10:35 Red Flag Arming
    - 10:45 Evac to Launch Control
    - 10:48 First Launch of Solid and/or Fill Start
    - 10:50 Second Launch
    - 10:52 Third launch

- 10:53 Hybrid, Liquid, and Tribrid Fire Window closes any team not leaving the ground at 10:53 must abort
- 11:02 Finish abort
- 11:05 arrive to pull off rail and pack up
- 11:12 pad clear and removal of the last attempted team
- 11:15 next team(s) starts setup

Static Test Area

Static Probationary Levels

There will be three Probationary Levels (PL-n) within Statics. Rather than total Impulse, they are governed by the maximum allowable total volume of **Gaseous and Liquid Pressurants and Propellants (GLPP)** to minimize the potential yield of a catastrophic failure. This includes ullage, pressurant, liquid and gaseous oxidizer, and liquid and gaseous fuel. The volumes are set such that good performance matches the existing framework of impulses derived from propellant densities with historical ullages and pressurant volumes. Rather than the team over-building a tank and “whoops” into something larger, if the team extracts more performance from a finite unyielding volumetric limit, then they have done a good job and are allotted that extra impulse. They are also set to push teams to develop higher efficiency projects that have margin in performance so that a lower performance firing at competition can still yield a partially successful flight. Once a team passes a probationary level they are free to operate in the next so long as the team does not fail to successfully fire an engine that range of volumes for two years. By the second season without a successful firing they will be dropped down by at least 1 probationary level. Probation level stands until an engine fire exceeds to a Specific Impulse greater than the level’s isp target requirement. All teams are assumed to be in PL 1 until evidence is provided by the team that they meet a higher probationary limit. Teams must supply evidence of current effective probationary level by the first range safety report. Engine test stand data showing total impulse, propellant and pressurant masses for that test will be required. This will be matched with our records and verified. In the event that a deviation is granted conditionally (with deliverables, such as an intermediate scale test, due during the year) the team must have demonstrated they have fully met those conditions by the final range safety report, otherwise the team is required to obey the original probationary level at competition.

Probationary Level 1

- **(2 - 8 Liters of GLPP)**
- This is a teams first Hybrid, Liquid, or Tribrid Rockets in the Statics Category. For Hybrid, Liquid, and Tribrid engines we highly recommend a 2 Liter entry to spend limited funds on the fabrication and testing of a GSE system and extensive testing campaigns. This

will keep the test cost low giving you more tests per dollar building your team's understanding of your system. Continued use of this engine after your competition year will allow for subscale testing of methods of increasing efficiency without larger more costly tests.
- Engines must exceed 100s to remove PL 1 probationary limit and build an engine with a tank with greater than 8 Liters of GLPP.

Probationary Level 2

- **(8 - 20 Liters of GLPP)**
- Hybrid, Liquid, and Tribrid Engines must exceed 145s to remove PL 2 probationary limit and build an engine with tank volume greater than 20 Liters of GLPP.

Probationary Level 3

- **(20 - 32 Liters of GLPP)**
- Hybrid, Liquid, and Tribrid Engines must exceed 170s to remove PL 3 probationary limit and build an engine with greater than 32 Liters of GLPP.

Cleared Static Teams

Non Probationary Teams that have passed their probationary levels that perform lower than 170s from the static fire will be considered an unsuccessful firing and can result in a team being put back on probation if they are unsuccessful in firing by the next year's competition.

Statics Operations

Statics teams will demonstrate compliance with the Fill-to-Fire time requirements for advanced teams for the final range safety report, including a test report and supporting photo or video evidence of the hot fire or cold-flow test as part of the recording for your wet dress rehearsal. This includes all automated and manual abort operational modes. Simulated abort/test scrub operations shall be rehearsed to demonstrate a complete return to safe system state within 25 minutes of the planned firing time (e.g. Fill-to-Fire time + 25 minutes). The intent is to create familiarity with the decision not to proceed with a test/launch and quickly reset. A marked-up copy of the test procedures shall be included in the test report noting any deviations from the planned procedure, and should be included as part of all other documented tests as good practice.

Static Fill to Fire Times

Tanks under 24 L GLPP F2F times must be under 20 Minutes and should be under 10. Abort times must be under 10 minutes and should be under 5

Tanks over 24 L GLPP F2F times must be under 30 Minutes and should be under 15. Abort times must be under 15 minutes and should be under 8

Office hours tentative

Peter all times EST
Mo 11-12AM LC
Tu 11-12AM FAROUT
W 11-12AM LC
Th 11-12AM LC
Sa 8AM - 9AM FAROUT
Su 11-12AM LC

Shane’s office hours to follow